Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/01/2004
PI Name:	Ma, Nancy Ph.D.		
Project Title:	Models of Mass Transport during Microgravity Crystal Growth of Alloyed Semiconductors in a Magnetic Field		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-7919	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/01/2001	**End Date:**	11/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Charles Rosenblatt		
COI Name (Institution):			
Grant/Contract No.:	NAG8-1817		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This grant involves two major tasks. The first involves the development of models which will treat the convection driven by compositional variations in the terrestrial vertical Bridgman process with an axial magnetic field in order to predict the composition of the entire alloyed crystal. These models would be validated by results from experiments which have been conducted at NASA MSFC. The second task will involve extending the terrestrial models to treat magnetically-damped compositionally-driven buoyant convection during Bridgman growth in space. The objective of the proposed research is to develop models which investigate the effects of the microgravity environment on segregation in alloyed semiconductor crystals. The significance of the results of this research is two-fold. First, these investigations would contribute to the fundamental physical understanding of the effects of terrestrial and microgravity as well as magnetic fields on compositionally-driven buoyant convection. Second, these models would guide process optimization for the production of alloyed semiconductor crystals such as GeSi, HgCdTe and PbTnTe, which are very important materials which serve as bases for optoelectronic devices.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	To date, models which have been developed treat dopant transport with magnetic fields in which convective dopant transport is driven by thermal buoyant convection both on earth and in space. Models which have investigated the effects of solutal convection on the species distribution in alloyed semiconductor crystals grown in magnetic fields on earth have been developed and two journal articles have appeared in print and one journal article is in press.
Bibliography Type:	Description: (Last Updated: 03/27/2006)
Abstracts for Journals and Proceedings	Nancy Ma, John S. Walker and Laurent M. Witkowski "Thermocapillary instability with a rotating magnetic field and system rotation" ASME Proceedings of the International Mechanical Engineering Congress and Exposition, Washington D. C., pp. 297-303, November 2003 pp. 297-303 , Jan-2004
Articles in Peer-reviewed Journals	Nancy Ma, John S. Walker and Laurent M. Witkowski "Combined effects of rotating magnetic field and rotating system on the thermocapillary instability in the floating zone crystal growth process" ASME Journal of Heat Transfer, vol. 126, no. 2, pp. 230-235, April 2004 , Apr-2004
Articles in Peer-reviewed Journals	Martin V. Farrell and Nancy Ma "Macrosegregation during alloyed semiconductor crystal growth in strong axial and transverse magnetic fields" International Journal of Heat and Mass Transfer, vol. 47, no. 14/16, pp. 3047-3055, July 2004 , Jul-2004
Presentation	Martin V. Farrell, Nancy Ma "Alloy segregation during Bridgman crystal growth in the presence of axial and transverse magnetic fields" AACG 15th American Conference on Crystal Growth and Epitaxy, Keystone, Colorado Jul-2003